

# Amirhossein Torabi

SOFTWARE ENGINEER

Turin, Italy

☎ +39 351 924 9022 | ✉ amirtorabi.a78@gmail.com | 🏠 amirhosseintorabi.github.io | 📺 AmirhosseinTorabi | 🌐 amirhosseintorabi

**Summary** — Full-stack **Software Engineer** with hands-on experience building scalable, production-grade web applications. Proficient in **TypeScript, React, Next.js**, and **Node.js/Express**, with a strong focus on **clean architecture, REST API design**, and **maintainable code**. Experienced in delivering systems across the full SDLC — from UML design and CI/CD pipelines to containerised deployment. M.Sc. thesis research focuses on evaluating **SOLID principles for LLM-driven software maintenance**, combining software engineering rigour with applied AI analysis.

**Core Competencies** — Full-Stack Development • REST API Design • Software Architecture • CI/CD • Agile/Scrum • Testing & TDD • Database Design • Containerisation

## Skills & Languages

**Frontend** HTML, CSS, JavaScript, TypeScript, React, Next.js, Tailwind CSS  
**Backend** Node.js, Express, REST APIs, Socket.io, NextAuth, JWT, Passport.js, bcrypt  
**Databases** PostgreSQL, SQLite, Prisma ORM, Sequelize  
**Testing** Jest, React Testing Library, Supertest — unit, integration & e2e  
**Design** UML, Use-Case Modeling, System Design, Deployment Diagrams  
**DevOps** Docker, GitHub Actions (CI/CD), Git, Postman  
**Languages** English (C1), Persian (Native), Italian (A2)

## Experience

### HUB Company

Web Developer / Webmaster

Shiraz, Iran

Feb 2021 – Sep 2023

- Owned end-to-end development and maintenance of the company's production website, managing uptime, **security hardening**, and plugin integrations.
- Improved organic search visibility by restructuring content and applying **SEO best practices**, resulting in measurable improvements in page indexing and ranking.
- Optimised site performance through **caching configuration** and asset management, reducing load times and improving Core Web Vitals scores.

### Kimia College (Sharif Financial Institute) – Quantum Incubator

Intern

Iran

Sep 2020 – Dec 2021

- Supported coordination of **8+ early-stage startup teams** across structured incubation programmes, tracking milestones and facilitating **20+ mentorship and training sessions**.

## Education

### Politecnico di Torino

M.Sc. in Computer Engineering – Software Engineering

Turin, Italy

Oct 2023 – Sep 2026 (Expected)

- Thesis: *Evaluating SOLID Principles for LLM-Driven Software Maintenance*

### Islamic Azad University, Shiraz Branch

B.Sc. in Computer Engineering – Software Engineering

Shiraz, Iran

Sep 2017 – Jul 2022

## Main Projects

### Participium

Politecnico di Torino | Nov 2025 – Jan 2026

- Built a full-stack civic reporting platform (team of 6) using **Next.js, Prisma ORM**, and **PostgreSQL**, following a 3-layer Controller–Service–Repository architecture across 10 database models.
- Implemented real-time communication via **Socket.io** and a **Telegram Bot** supporting report creation, status tracking, and push notifications.
- Delivered **108 test files** (87 unit + 21 integration) using **Jest** and **React Testing Library**, automated via **4 GitHub Actions CI/CD pipelines** with Codecov coverage reporting.
- Designed a multi-role access system (citizen, officer, maintainer, admin, external company) with **NextAuth** authentication across 6 user-facing interfaces.
- Tech Stack:** TypeScript, React, Next.js, Tailwind CSS, Prisma ORM, PostgreSQL, Socket.io, Jest, GitHub Actions, Docker

### EZElectronics

Politecnico di Torino | Apr 2024 – Jul 2024

- Designed a full-stack e-commerce system (team of 4) from **UML use-case diagrams** through to a **TypeScript/Express**

REST API with 5 resource routers (users, sessions, products, carts, reviews).

- Delivered a **495-test-case** suite spanning unit, DAO, controller, route, and integration layers using **Jest** and **Supertest** (white-box statement coverage).
- Implemented session-based authentication via **Passport.js** with role-based access control; containerised with **Docker**.
- **Tech Stack:** TypeScript, React, Node.js, Express, SQLite, Passport.js, Jest, Supertest, Docker

### Social Budget

*Politecnico di Torino | Jun 2024 – Jul 2024*

- Built a participatory budgeting web app with a **React/Vite** frontend and a **Node.js/Express** REST API (15+ endpoints across 5 routers) backed by **Sequelize ORM** and **SQLite**.
- Implemented a 4-phase workflow (setup → proposal → voting → results) with **JWT** authentication, **bcrypt** password hashing, and role-based access control.
- **Tech Stack:** JavaScript, React, Bootstrap, Node.js, Express, Sequelize, SQLite, JWT

*More projects available on my GitHub: [github.com/AmirhosseinTorabi](https://github.com/AmirhosseinTorabi)*